The Mathematics of PageRank

Fan Chung

University of California, San Diego

Web Images Maps News Shopping Gmail more ▼

fan@gmail.com | My Notebooks | Web History | My Account | Sign out

Google

mathematicians Search Advanced Search Preferences

Web Personalized Results **1** - **10** of about **4,090,000** for **mathematicians** [definition]. (**0.23** seconds)

Mathematician - Wikipedia, the free encyclopedia
One of these problems, the Poincaré conjecture (1904), was proven by Russian **mathematician** Grigori Perelman in a paper released in 2003; peer review was **...**
en.wikipedia.org/wiki/Mathematician - 37k - Cached - Similar pages - Note this

Mathematician Biography Index
A site giving brief biographies for virtually every important **mathematician** in history.
www-groups.dcs.st-and.ac.uk/~history/BiogIndex.html - 6k -
Cached - Similar pages - Note this

MySpace.com - **Mathematicians** - The Tri County Region of Upstate **...**
MySpace music profile for **Mathematicians** with tour dates, songs, videos, pictures, blogs, band information, downloads and more.
www.myspace.com/**mathematicians** - 186k - Cached - Similar pages - Note this

Mathematicians
Mathematicians use mathematical theory, computational techniques, algorithms, and the latest computer technology to solve economic, scientific, engineering, **...**
www.bls.gov/oco/ocos043.htm - 54k - Cached - Similar pages - Note this

Mathematicians

PageRank

Random walks

Spectral graph theory

Concepts from spectral geometry

Probabilistic methods

Combinatorial algorithms

Graph theory

Mathematics

Outline of the talk

- Motivation
- Define PageRank
- Local cuts and the `goodness' of a cut
- The "geometry" of a network

  -- mathematical analysis using:

  - eigenvectors
  - random walks
  - PageRank
  - heat kernel

Search Engines:







眾裡尋他千百度
驀然回首　那人卻在
燈火闌珊處

辛棄疾　青玉案

Ask, and it will be given to you; seek, and you will find; knock, and it will be opened to you.

Mathew 7:7

Google

Our Search: Google Technology

Home

About Google

Help Central

Google Features

Services & Tools

Our Technology

- **Why Use Google**

Benefits of Google

Find on this site:

Search

Google searches more sites more quickly, delivering the most relevant results.

Introduction

Google runs on a unique combination of advanced hardware and software. The speed you experience can be attributed in part to the efficiency of our search algorithm and partly to the thousands of low cost PC's we've networked together to create a superfast search engine.

The heart of our software is PageRank™, a system for ranking web pages developed by our foundersLarry Page and Sergey Brin at Stanford University. And while we have dozens of engineers working to improve every aspect of Google on a daily basis, PageRank continues to play a central role in many of our web search tools.

PageRank Explained

PageRank relies on the uniquely democratic nature of the web by using its vast link structure as an indicator of an individual page's value. In essence, Google interprets a link from page A to page B as a vote, by page A, for page B. But, Google looks at considerably more than the sheer volume of votes, or links a page receives; for example, it also analyzes the page that casts the vote. Votes cast by pages that are themselves "important" weigh more heavily and help to

What is PageRank?

Google's answer:

PageRank Explained

PageRank relies on the uniquely democratic nature of the web by using its vast link structure as an indicator of an individual page's value. In essence, Google interprets a link from page A to page B as a vote, by page A, for page B. But, Google looks at considerably more than the sheer volume of votes, or links a page receives; for example, it also analyzes the page that casts the vote. Votes cast by pages that are themselves "important" weigh more heavily and help to make other pages "important." Using these and other factors, Google provides its views on pages' relative importance.

What is PageRank?

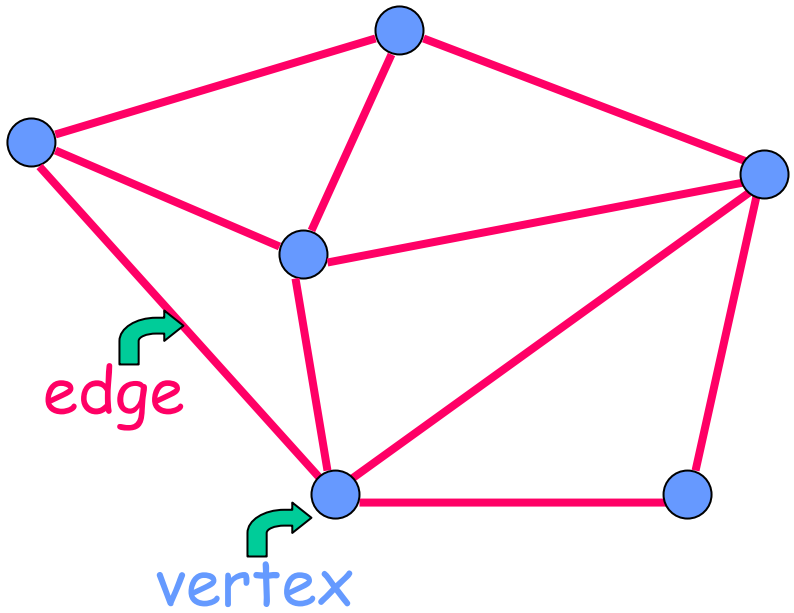
PageRank is a well-defined operator on any given graph, introduced by Sergey Brin and Larry Page of Google in a paper of 1998.



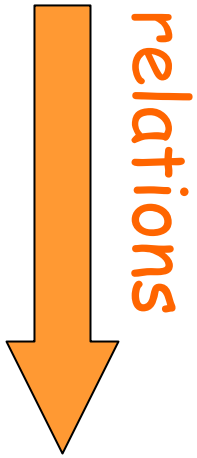
Graph models

Vertices	Edges
cities	flights
people	pairs of friends
telephones	phone calls
web pages	linkings
genes	regulatory effect

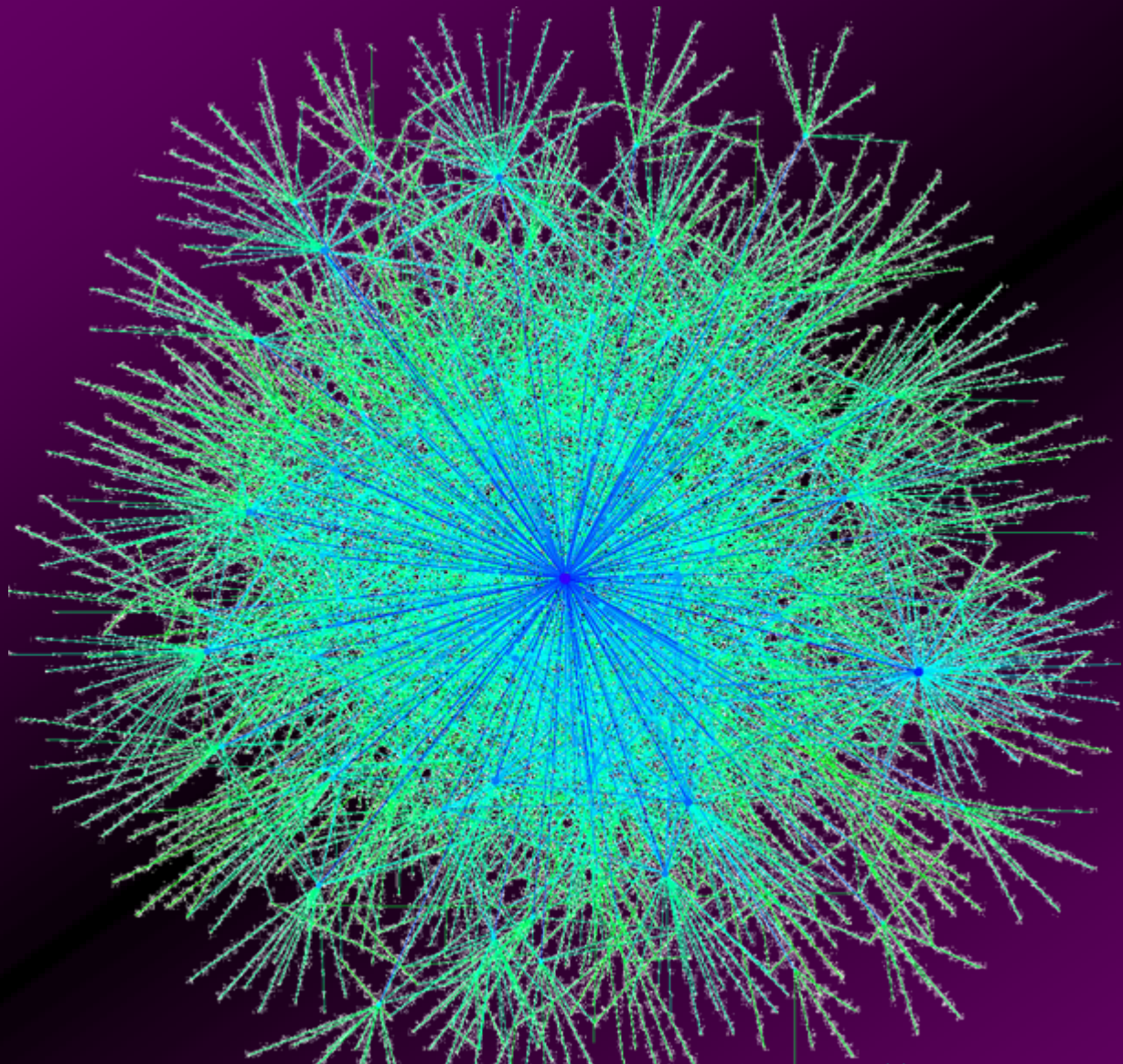
A graph G = (V,E)



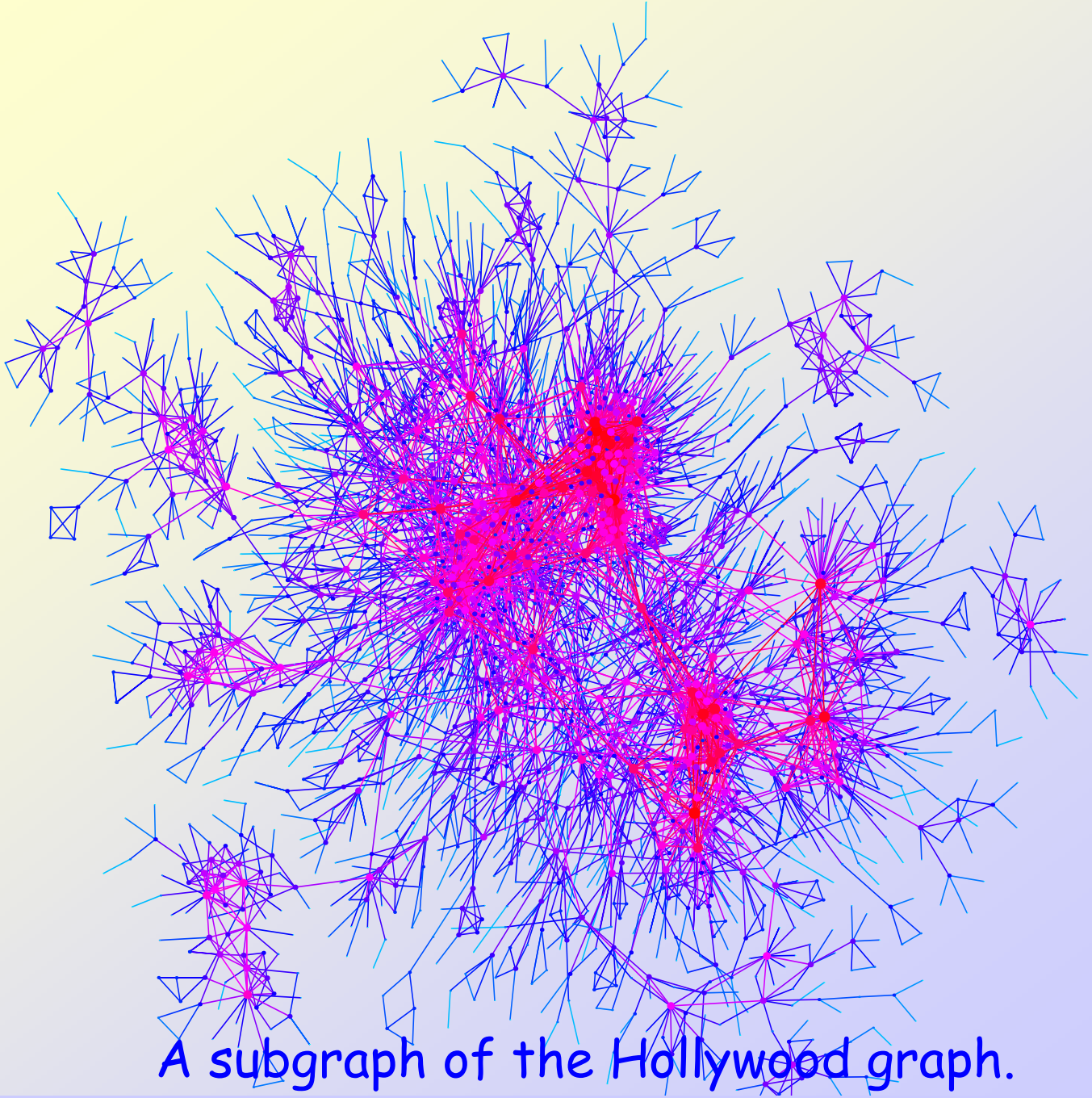
Information



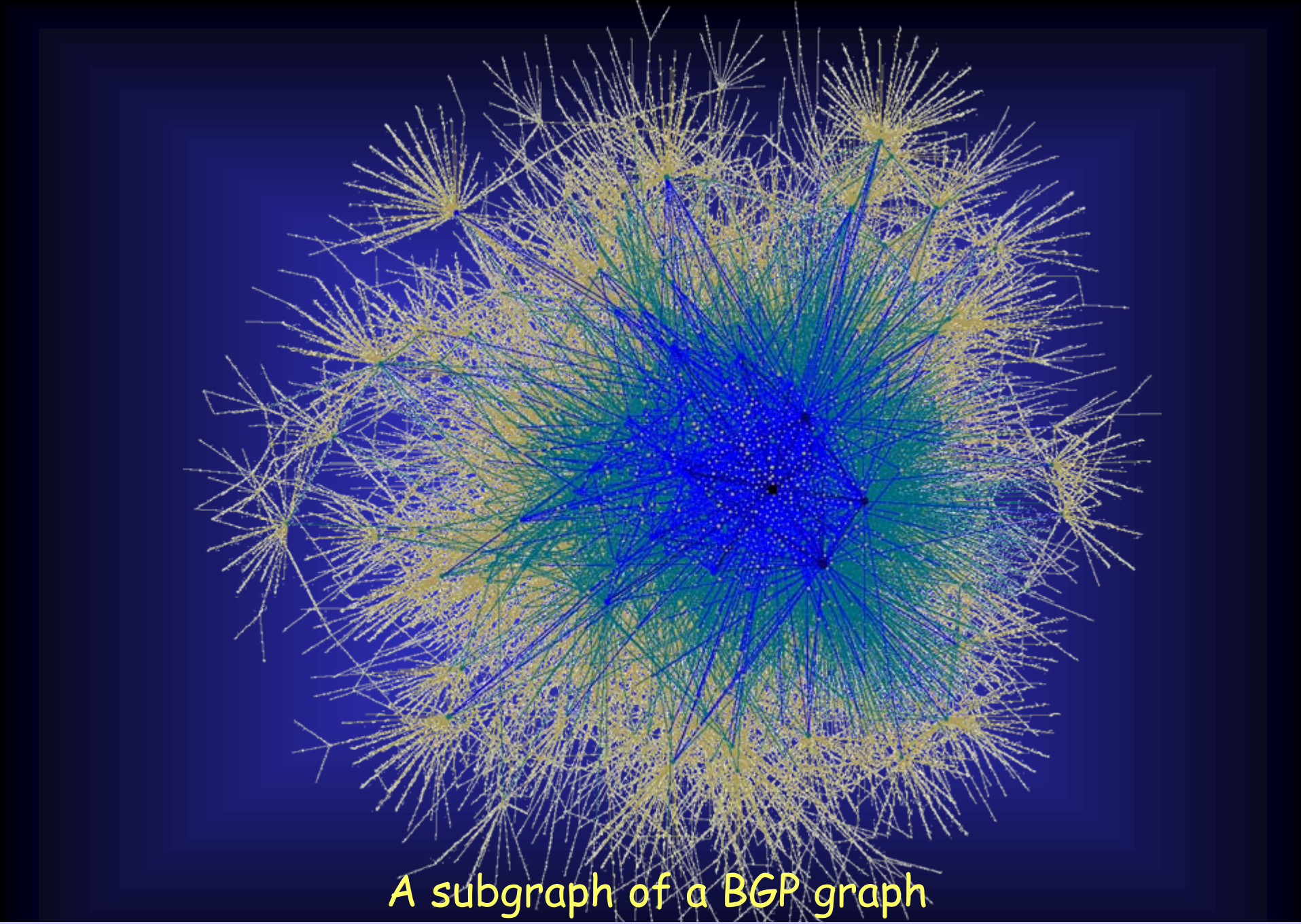
Information network



An induced subgraph of the collaboration graph with authors of Erdös number ≤ 2.

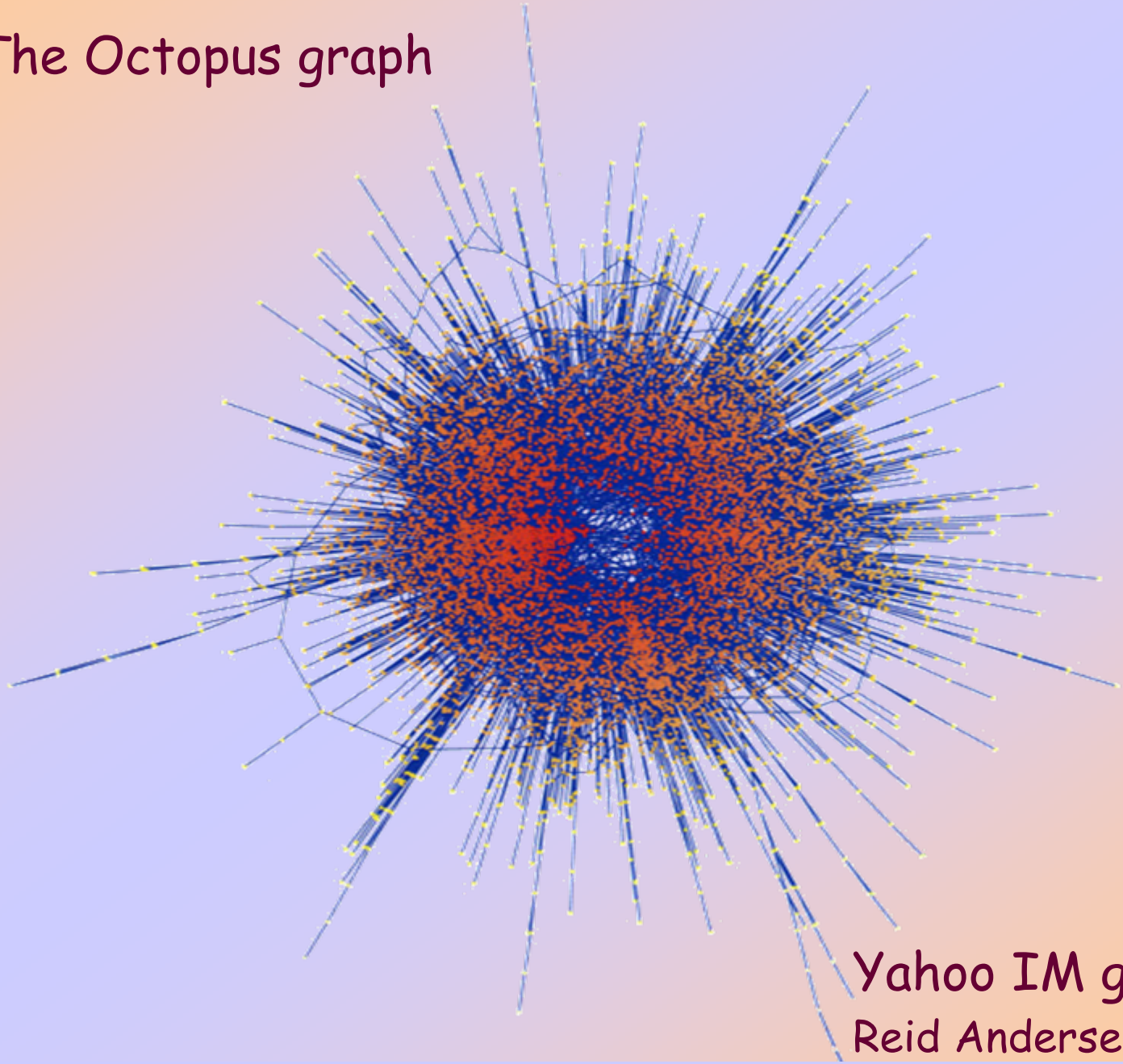


A subgraph of the Hollywood graph.



A subgraph of a BGP graph

The Octopus graph



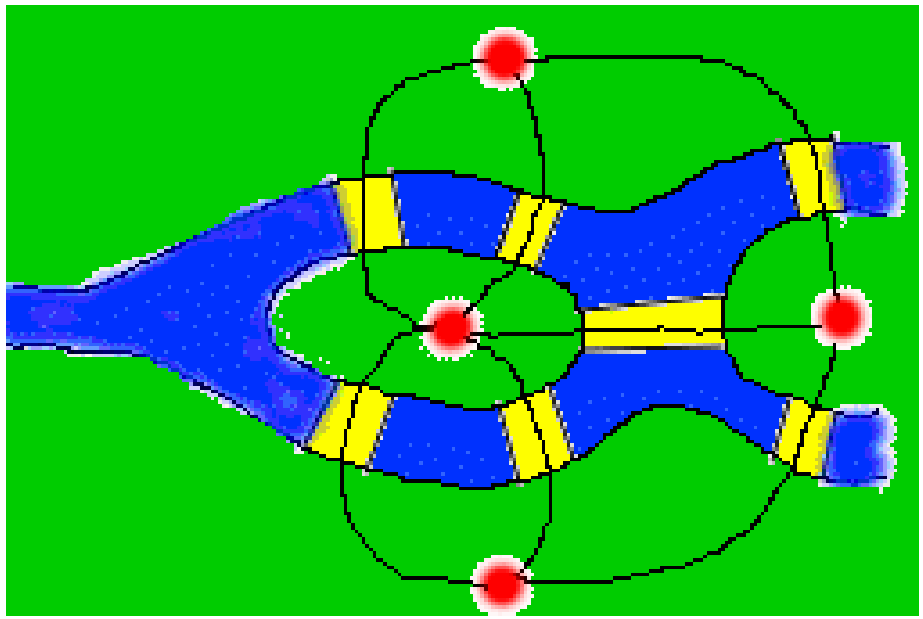
Yahoo IM graph

Reid Andersen 2005

Graph Theory has 250 years of history.

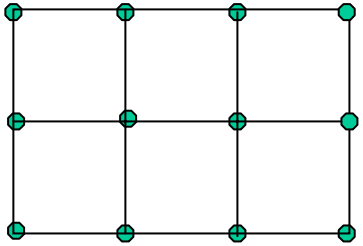


Leonhard Euler
1707-1783

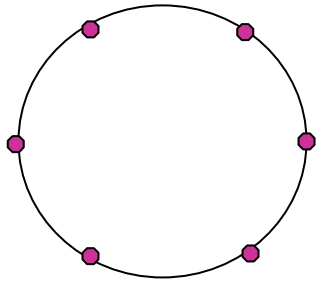


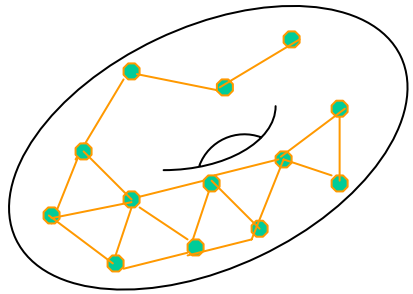
The Bridges of Königsburg
Is it possible to walk over every bridge once and only once?

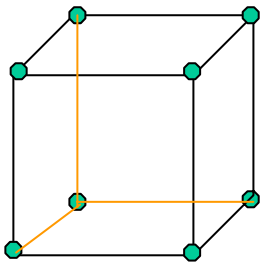
Geometric graphs

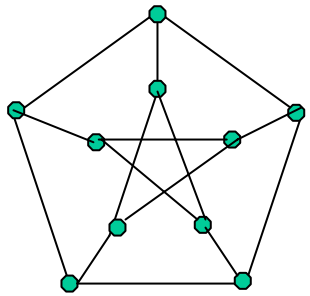


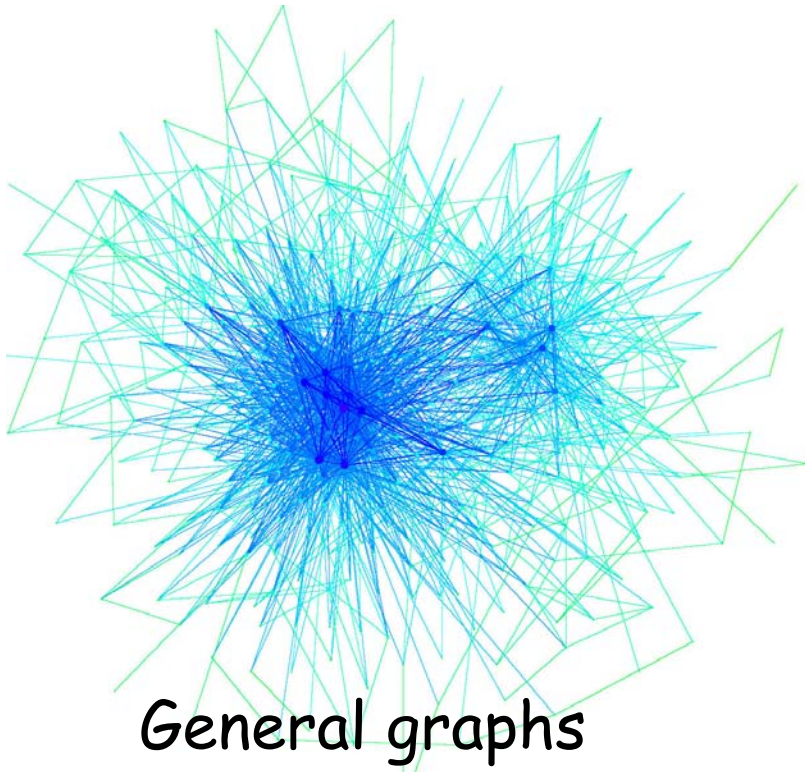
Topological graphs









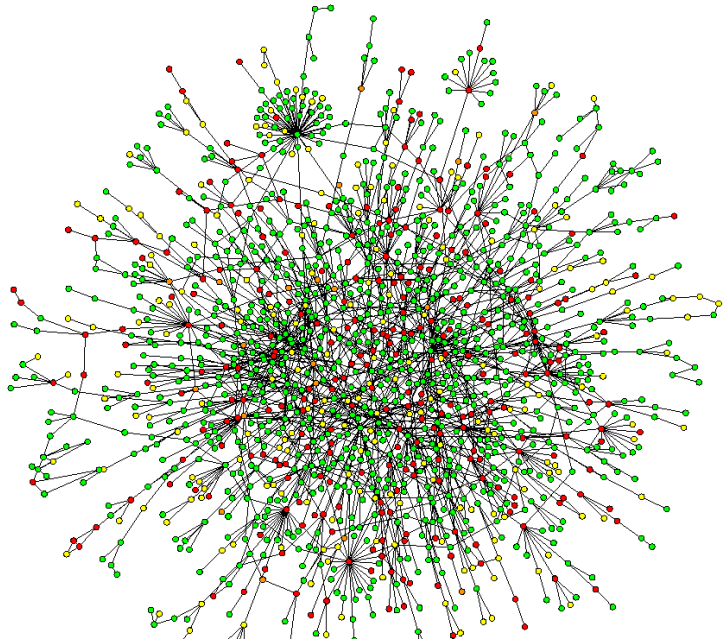


Algebraic graphs

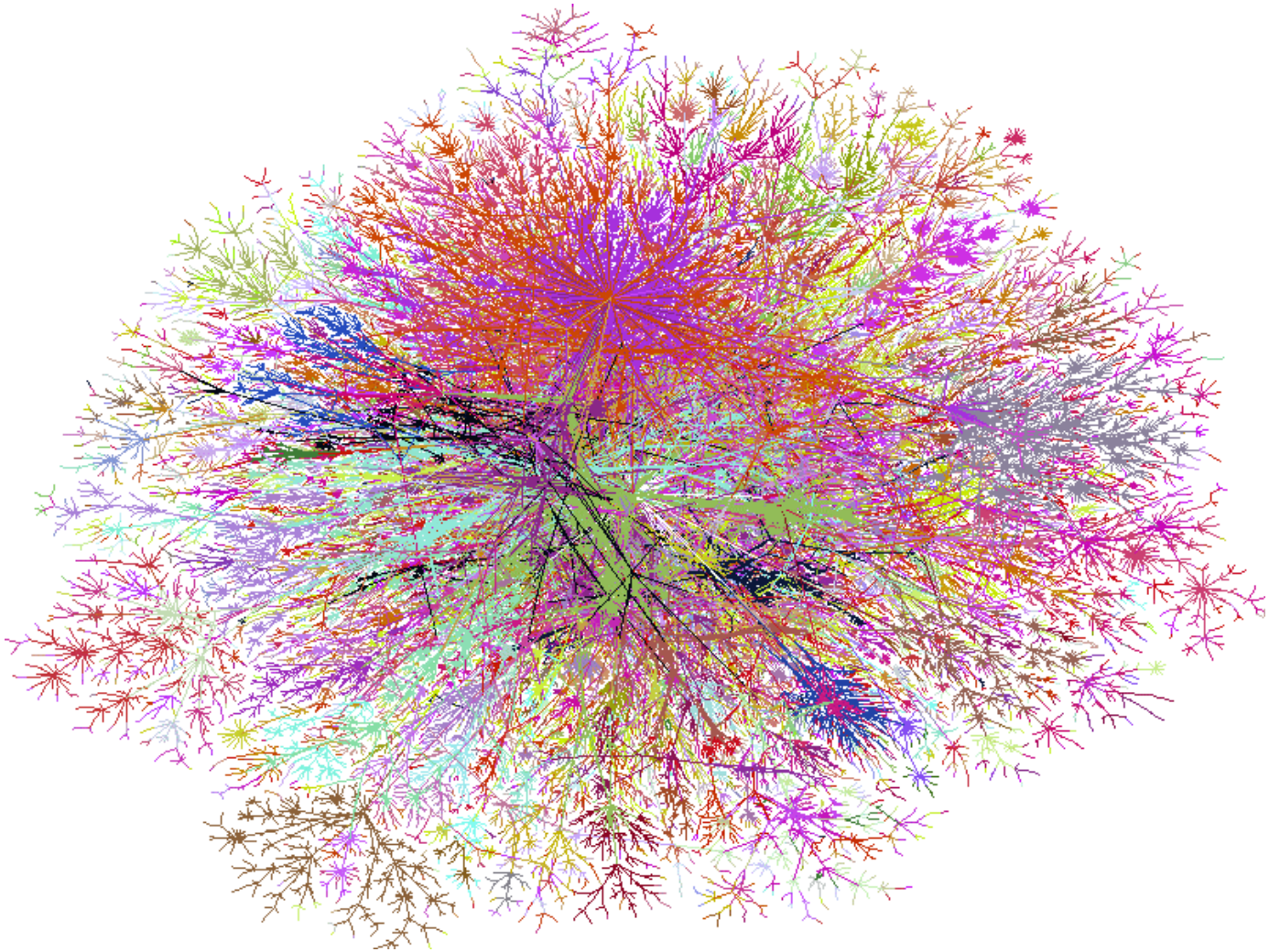
General graphs

Massive data → Massive graphs

- WWW-graphs
- Call graphs
- Acquaintance graphs
- Graphs from any data base



protein interaction network
Jawoong Jeong



Big and bigger graphs ⇒ New directions.

Many basic questions:

- **Correlation among vertices?**
- **The `geometry' of a network ?**

  distance, flow, cut, …
- **Quantitative analysis?**

  eigenvalues, rapid mixing, …
- **Local versus global?**

A measure for the "importance" of a website

$$x_1 = R(x_{14} + x_{79} + x_{785})$$

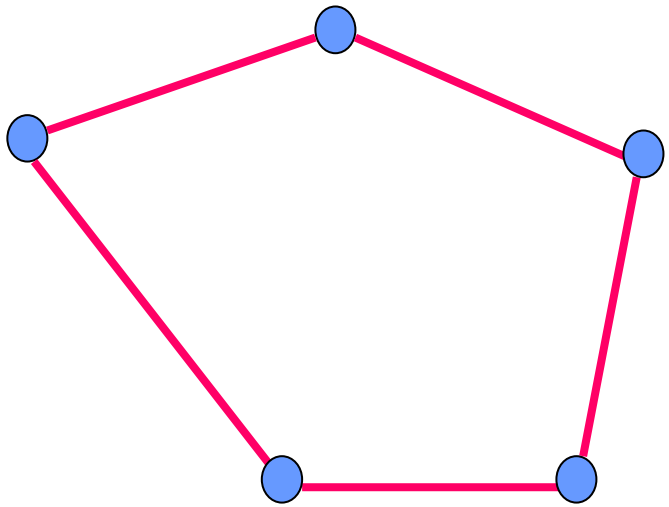
$$x_2 = R(x_{1002} + x_{3225} + x_{9883} + x_{30027})$$

$$\cdots \quad = \quad \cdots$$

The "importance" of a website is proportional to the sum of the importance of all the sites that link to it.

Adjacency matrix of a graph

Example: Adjacency matrix of a 5-cycle



$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix}$$

A solution for the "importance" of a website

$$x_1 = \rho(x_{14} + x_{79} + x_{785})$$

$$x_2 = \rho(x_{1002} + x_{3225} + x_{9883} + x_{30027})$$

$$\cdots \quad = \quad \cdots$$

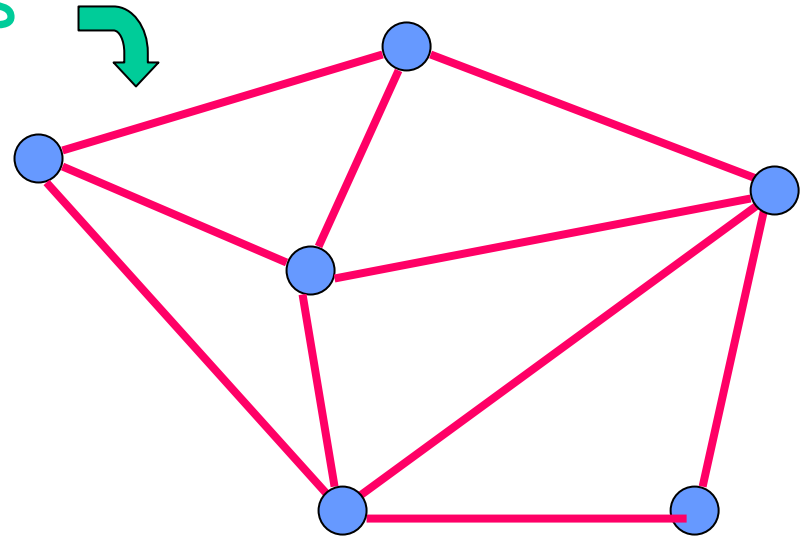
Solve $x_i = \rho \sum_{j=1}^{n} a_{ij} x_j$ for $\boldsymbol{x} = (x_1, x_2, \cdots, x_n)$



$\boldsymbol{x} = \rho A \boldsymbol{x}$ $\qquad A = \left[a_{ij} \right]_{n \times n}$

Graph models

(undirected) graphs



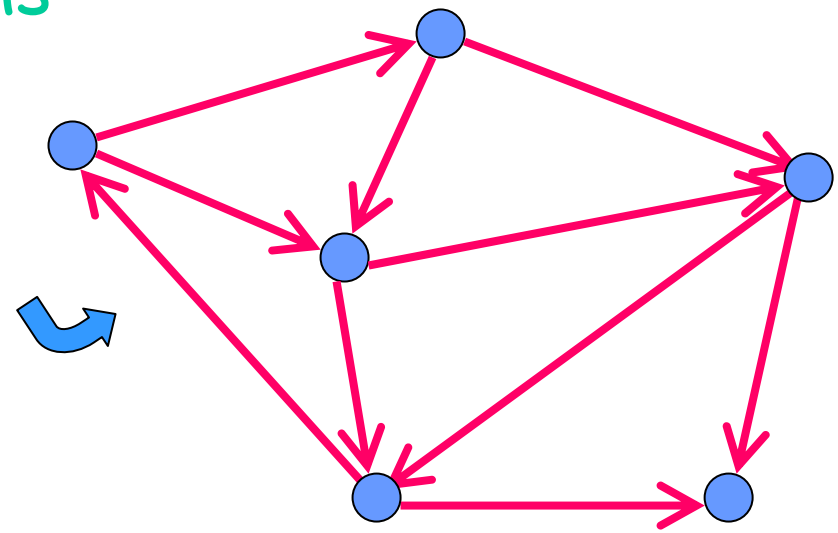
directed graphs

weighted graphs

Graph models

(undirected) graphs

directed graphs



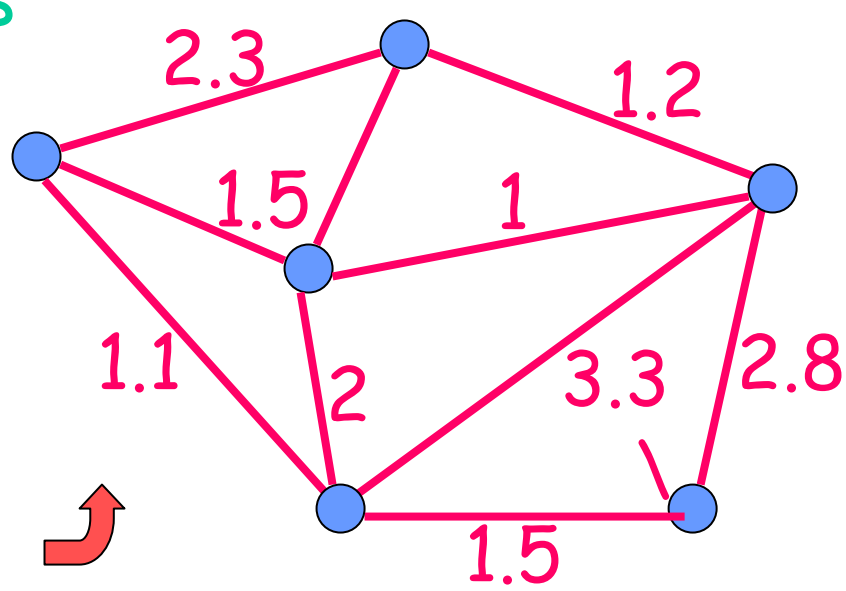
weighted graphs

Graph models

(undirected) graphs

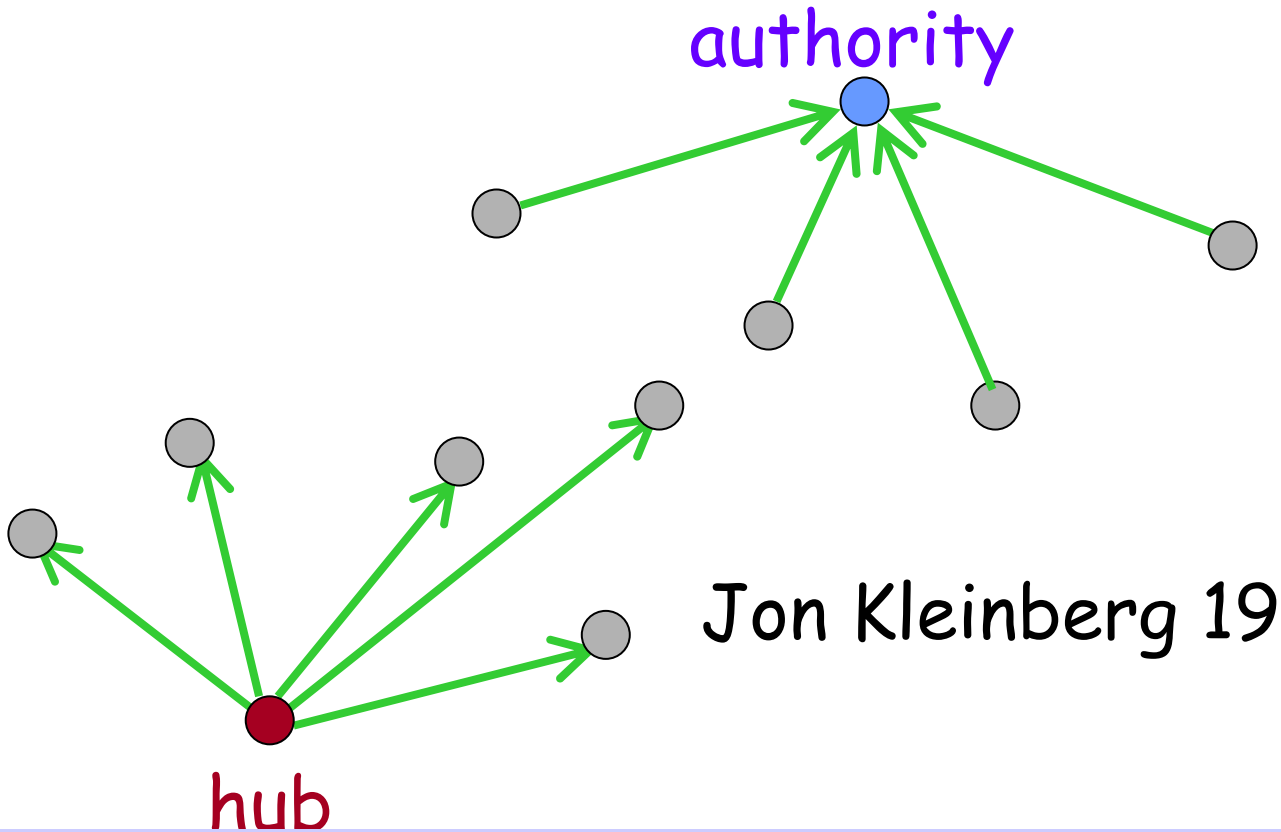
directed graphs

weighted graphs



In a directed graph,

there are two types of "importance":



Jon Kleinberg 1998

Two types of the "importance" of a website

Importance as Authorities : $\boldsymbol{x} = (x_1, x_2, \cdots, x_n)$

Importance as Hubs : $\boldsymbol{y} = (y_1, y_2, \cdots, y_m)$

Solve $\boldsymbol{x} = r\, A\, \boldsymbol{y}$ and $\boldsymbol{y} = s\, A^T \boldsymbol{x}$

$$\Rightarrow \begin{cases} \boldsymbol{x} = rs\, A\, A^T \boldsymbol{x} \\ \boldsymbol{y} = rs\, A^T A\, \boldsymbol{y} \end{cases}$$

Singular eigenvalue problems!

Eigenvalue problem for n x n matrix:.

$n \approx$ 30 billion websites

Hard to compute eigenvalues

Even harder to compute eigenvectors

In the old days,

compute for a given (whole) graph.

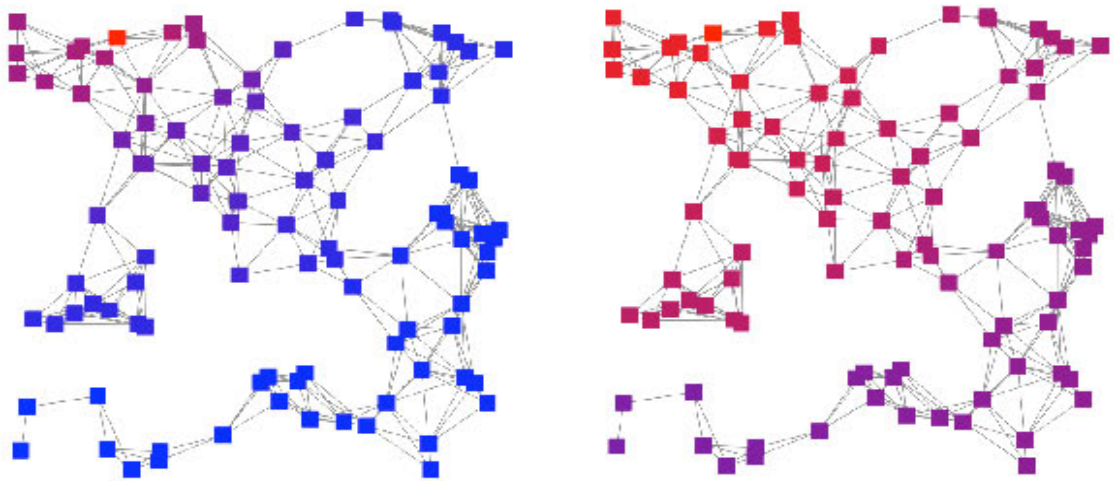
In reality,

can only afford to compute "locally".

The definition of PageRank given by

Brin and Page is based on

random walks.



Random walks in a graph.

G : a graph

P : transition probability matrix

$$P(u,v) = \begin{cases} \dfrac{1}{d_u} & \text{if } u : v, \\ 0 & \text{otherwise.} \end{cases}$$

d_u := the degree of u.

A lazy walk: $$W = \frac{I + P}{2}$$

Original definition of PageRank

A (bored) surfer

- either surf a random webpage

  with probability α

- or surf a linked webpage

  with probability $1-\alpha$



α : the jumping constant

$$p = \alpha(\tfrac{1}{n}, \tfrac{1}{n}, ..., \tfrac{1}{n}) + (1-\alpha)pW$$

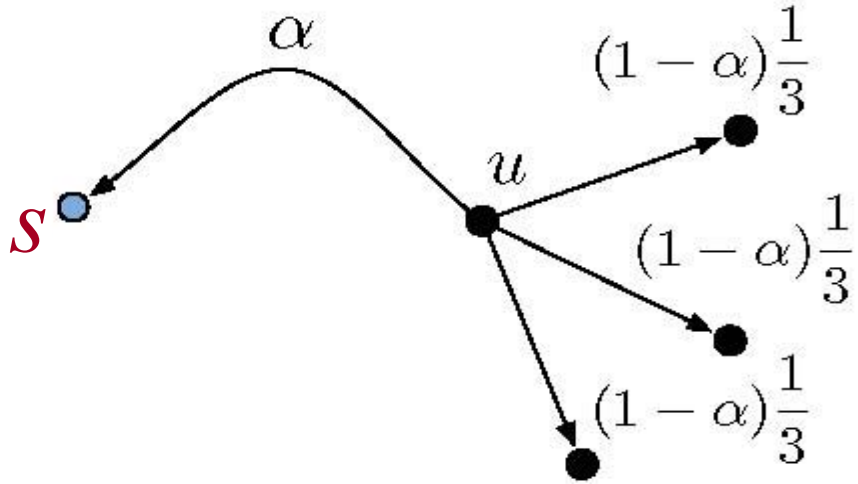
Definition of personalized PageRank

Two equivalent ways to define PageRank $pr(\alpha,s)$

(1) $$p = \alpha s + (1-\alpha) pW$$

s: the seed as a row vector

α : the jumping constant



Definition of PageRank

Two equivalent ways to define PageRank $p=pr(\alpha,s)$

(1) $$p = \alpha s + (1-\alpha) pW$$

(2) $$p = \alpha \sum_{t=0}^{\infty} (1-\alpha)^t (sW^t)$$

$s = (\frac{1}{n}, \frac{1}{n}, ..., \frac{1}{n})$ ⟹ the (original) PageRank

$s =$ some "seed", e.g., $(1, 0, ..., 0)$

⟹ personalized PageRank

How good is PageRank as a measure of correlationship?

~~Depends on the applications?~~

Rely on mathematics

both old and new.

Isoperimetric properties

"What is the shortest curve enclosing a unit area?"

In a graph G and an integer m,

what is the minimum cut disconnecting a subgraph of $\geq m$ vertices?



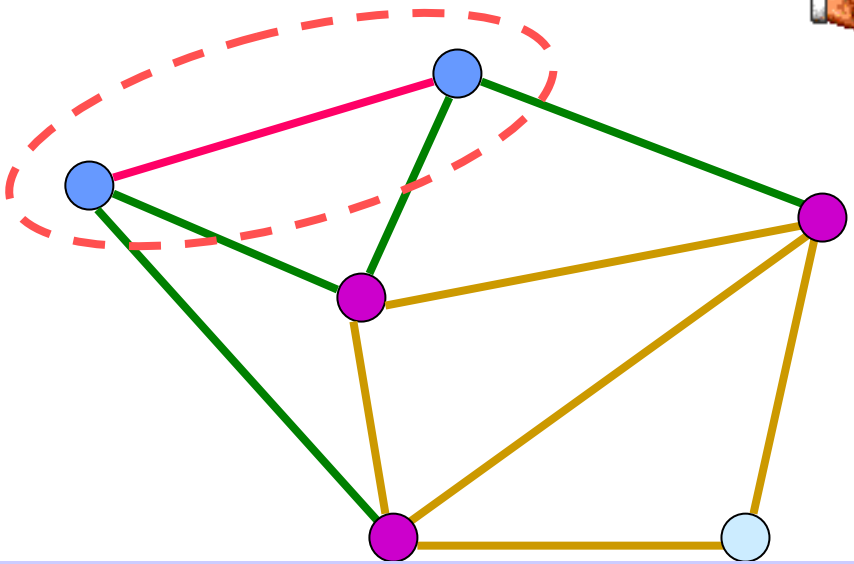
In a graph G, what is the minimum cut $e(S,V-S)$

so that $\dfrac{e(S,V-S)}{\mathrm{Vol}\ S}$ is the smallest?

How "good" is the cut?

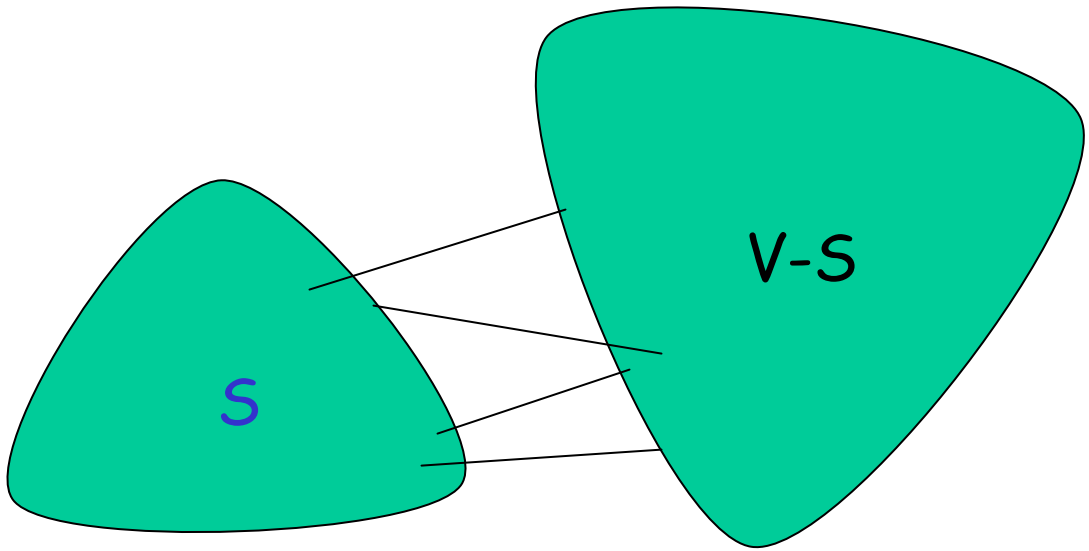
Two types of cuts:

- Vertex cut
- edge cut



$$\frac{e(S,V-S)}{\mathrm{Vol}\ S} \quad \Longleftrightarrow \quad \frac{e(S,V-S)}{|S|}$$

$$\mathrm{Vol}\ S = \sum_{v \in S} \deg(v) \qquad |S| = \sum_{v \in S} 1$$



The Cheeger constant for graphs

The Cheeger constant

$$\Phi_G = \min_S \frac{e(S,\bar{S})}{\min(vol\ S, vol\ \bar{S})}$$

The volume of S is $vol(S) = \sum_{x \in S} d_x$

Φ_G and its variations are sometimes called "conductance", "isoperimetric number", …

The Cheeger inequality

The Cheeger constant

$$\Phi_G = \min_S \frac{e(S, \bar{S})}{\min(vol\ S, vol\ \bar{S})}$$

The Cheeger inequality

$$2\Phi_G \geq \lambda \geq \frac{\Phi_G^2}{2}$$

- : the first nontrivial eigenvalue of the (normalized) Laplacian.

The spectrum of a graph

- Adjacency matrix

Many ways to define the spectrum of a graph.



How are the eigenvalues related to properties of graphs?

The spectrum of a graph

- Adjacency matrix

- Combinatorial Laplacian

$$L = D - A$$

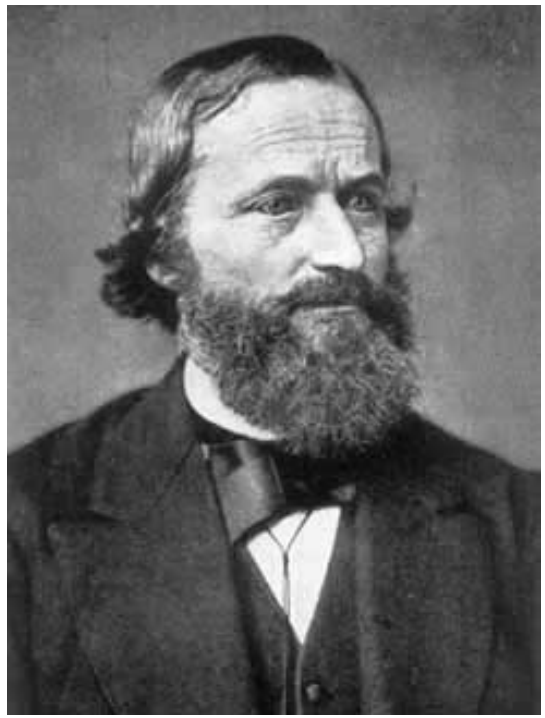
adjacency matrix

diagonal degree matrix

- Normalized Laplacian

Random walks

Rate of convergence



Gustav Robert Kirchhoff

1824-1887

The spectrum of a graph

Discrete Laplace operator

$$\Delta f(x) = \frac{1}{d_x} \sum_{y:\, x} (f(x) - f(y))$$

In a path P_n

$$= -\frac{1}{2}\{(f(x_{j+1}) - f(x_j)) - (f(x_j) - f(x_{j-1}))\}$$

$$-\frac{\partial^2}{\partial x^2} f(x) \leftrightarrow -\{\frac{\partial}{\partial x} f(x_{j+1}) - \frac{\partial}{\partial x} f(x_j)\}$$

The spectrum of a graph

loopless, simple

Discrete Laplace operator

$$\Delta f(x) = \frac{1}{d_x} \sum_{y:\, x} (f(x) - f(y))$$

$$L(x,y) = \begin{cases} 1 & if \quad x = y \\ -\dfrac{1}{d_x} & if \quad x \neq y \quad and \quad x : y \end{cases}$$

not symmetric in general



- Normalized Laplacian

symmetric
normalized

$$\mathrm{L}(x,y) = \begin{cases} 1 & if \quad x = y \\ -\dfrac{1}{\sqrt{d_x d_y}} & if \quad x \neq y \quad and \quad x : y \end{cases}$$

with eigenvalues

$$0 = \lambda_0 \leq \lambda_1 \leq \cdots \leq \lambda_{n-1} \leq 2$$

Can you hear the shape of a network?

λ dictates many properties of a graph.

- connectivity
- diameter
- isoperimetry (bottlenecks)
- … …

How "good" is the cut by using the eigenvalue λ ?

Finding a cut by a sweep

For $f : V(G) \to R,$ order the vertices

$$\frac{f(v_1)}{d_{v_1}} \geq \frac{f(v_2)}{d_{v_2}} \geq \cdots \geq \frac{f(v_n)}{d_{v_n}}.$$

Consider sets

$$S_j^f = \{v_1, \cdots, v_j\}, j = 1, \ldots, n,$$

and the Cheeger constant of S_j^f.

 Define

$$\Phi(f) = \min_j \Phi(S_j^f)$$

Finding a cut by a sweep

Using a sweep by the eigenvector, can reduce the exponential number of choices of subsets to a linear number.



Still, there is a lower bound guarantee by using the Cheeger inequality.

$$2\Phi \geq \lambda \geq \frac{\Phi^2}{2}$$

Four types of Cheeger inequalities.

Four proofs of Cheeger inequalities using:

- eigenvectors
- random walks
- PageRank
- heat kernel

Leading to four different

one-sweep partitioning algorithms.

Four proofs of Cheeger inequalities

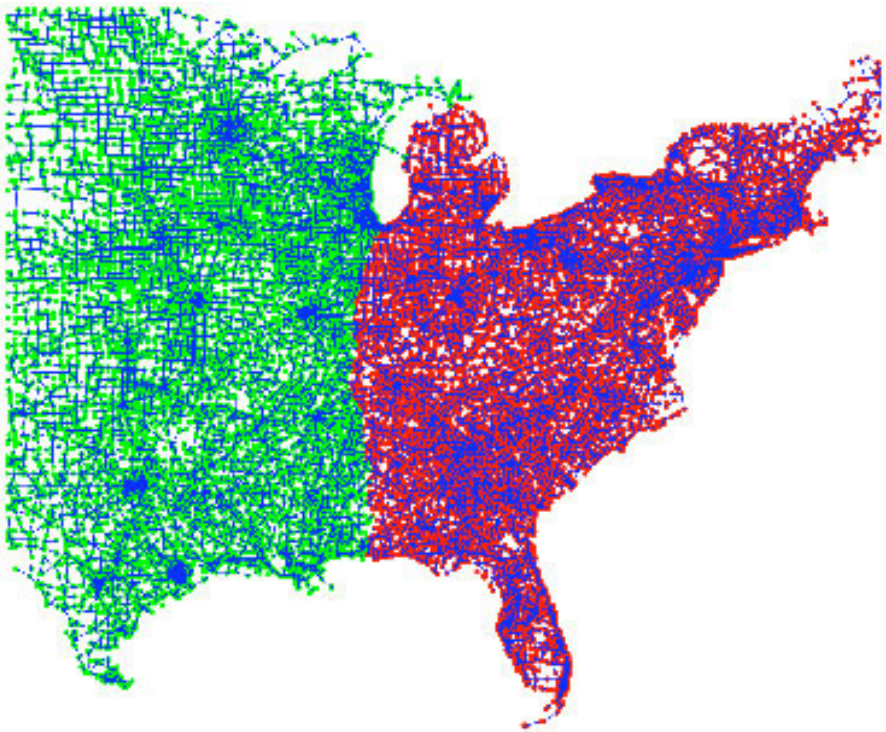
- graph spectral method 

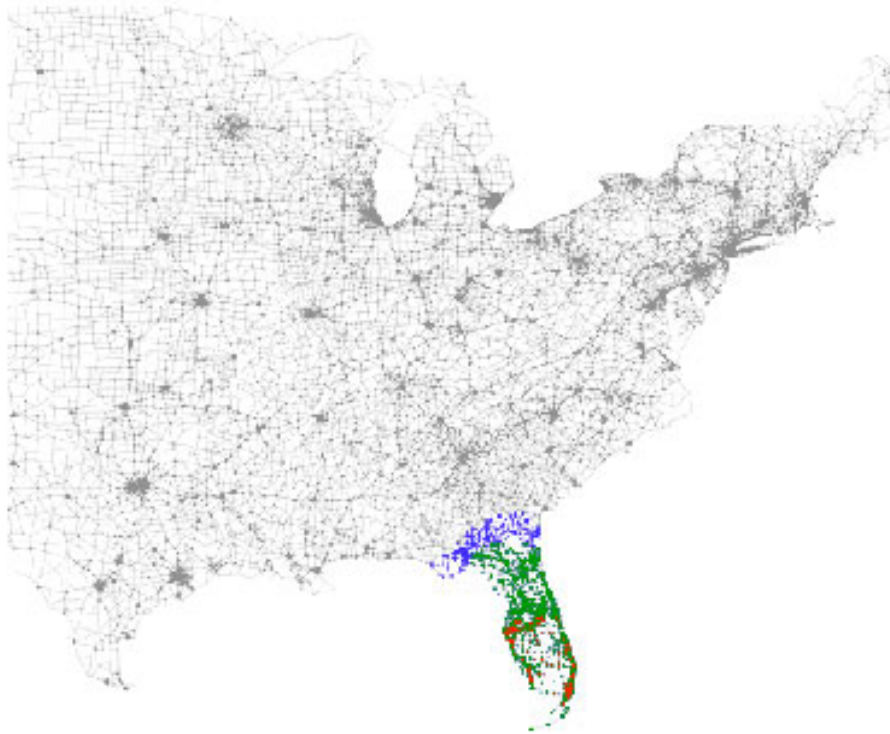
spectral partition algorithm

- random walks
- PageRank
- heat kernel

} local partition algorithms

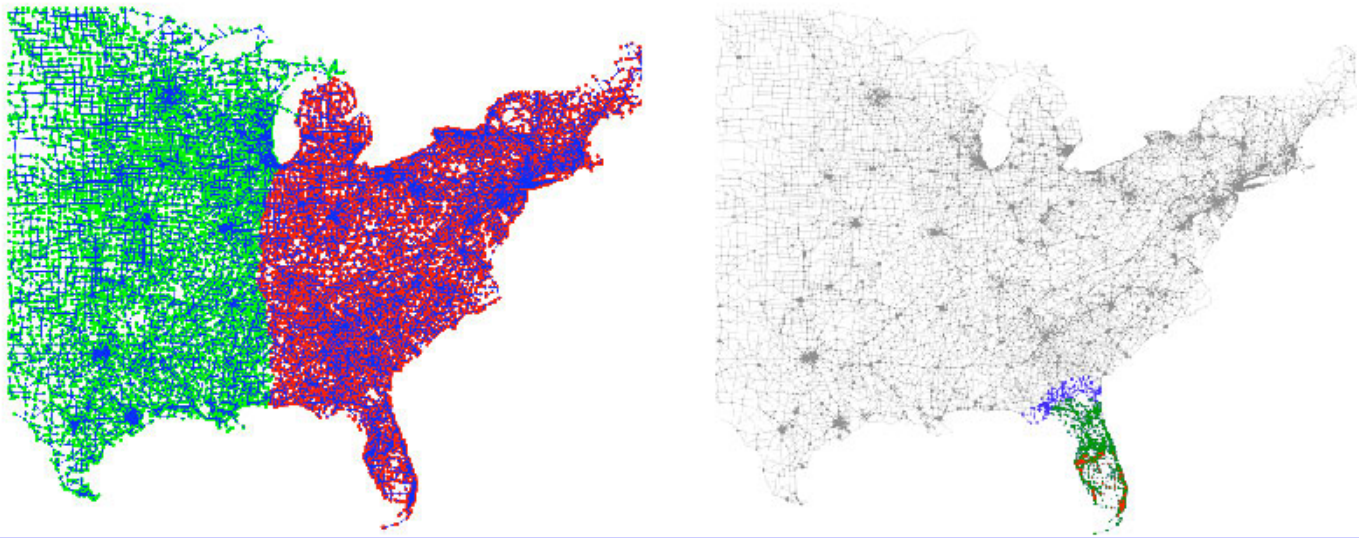
Graph partitioning ⟺ Local graph partitioning





What is a local graph partitioning algorithm?

A local graph partitioning algorithm finds a small cut near the given seed(s) with running time depending only on the size of the output.

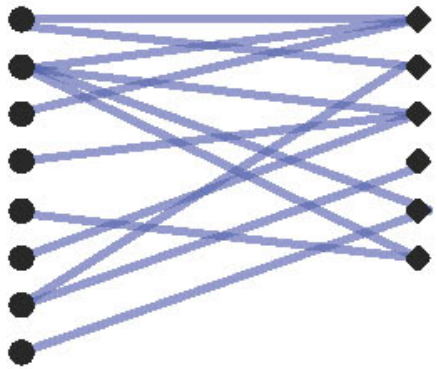


Two examples (Reid Andersen)

A bidding graph from Yahoo sponsored search

Phrases Advertiser IDs

e.g. Margarita Mix e.g. c8cbfd0bd74ba8cc



On the left are search phrases, on the right are advertisers.
Each edge represents a bid by an advertiser on a phrase.

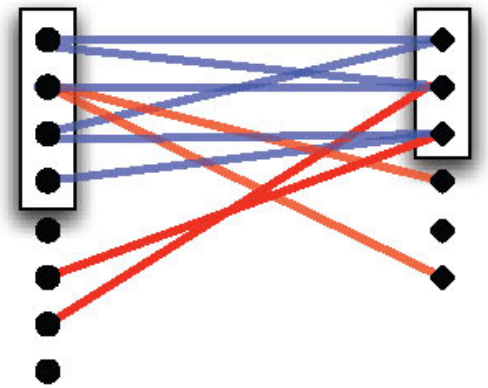
$400K$ phrases, $200K$ advertisers, and 2 million edges.

Two examples (Reid Andersen)

Submarkets in bidding graph

The bidding graph has submarkets, sets of bidders and phrases that interact mostly with each other.

Phrases about margarita mix Purveyors of margarita mix

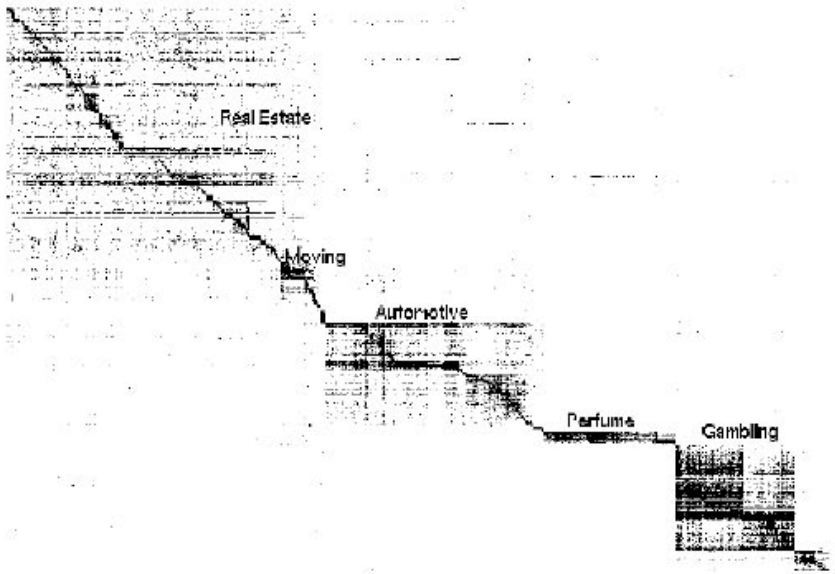


These sets of vertices (containing both advertisers and phrases) have small conductance.

Two examples (Reid Andersen)

Submarkets in the bigging graph

The bidding graph has numerous submarkets, related to real estate, flower delivery, hotels, gambling, ...



It is useful to identify these submarkets.

- Find groups of related phrases to suggest to advertisers.
- Find small submarkets for testing and experimentation.

Two examples (Reid Andersen)

Applying a local partitioning algorithm

Example in the sponsored search graph.

Starting with the seed vertex "alameda flower" and the target size 2000, our algorithm finds a set of 300 bidders and phrases related to flower stores in the San Francisco area. This set has small conductance.

alameda flower,	florist francisco in san,
alameda florist,	flower menlo park,
burlingame flower,	bruno flower san,
flower rafael san,	city flower redwood,
city daly flower,	florist rafael san, . . .

To find this submarket, our algorithm examined only 1200 vertices out of 653, 260 in the graph.

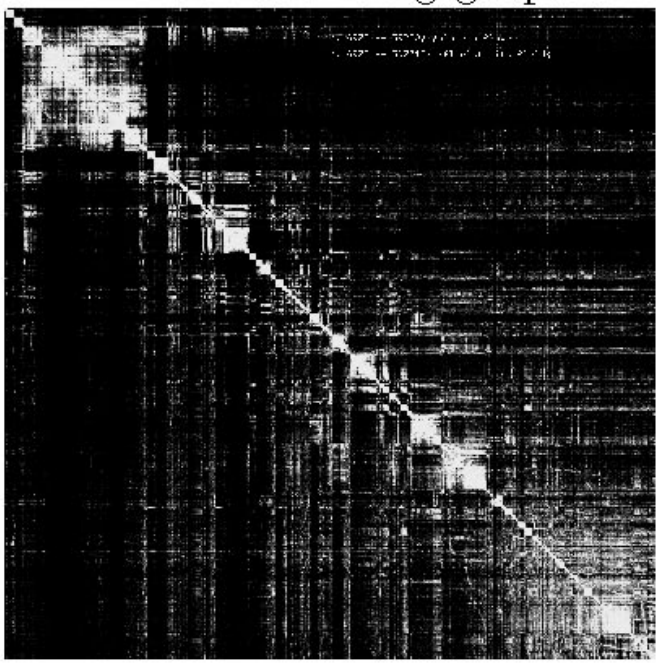
The time spent per vertex is roughly a constant, if the target

Two examples (Reid Andersen)

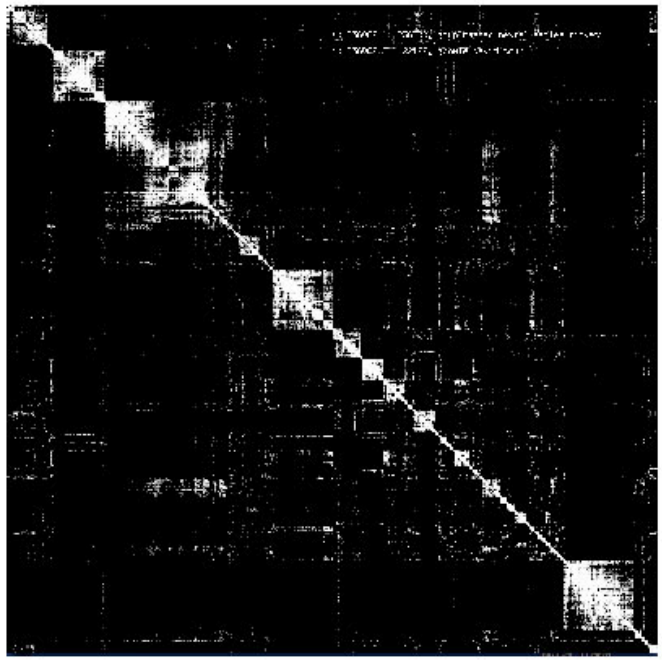
There are thousands of submarkets

If you want to decompose the graph into submarkets, the running time is determined by the time spent per vertex.

Bidding graph



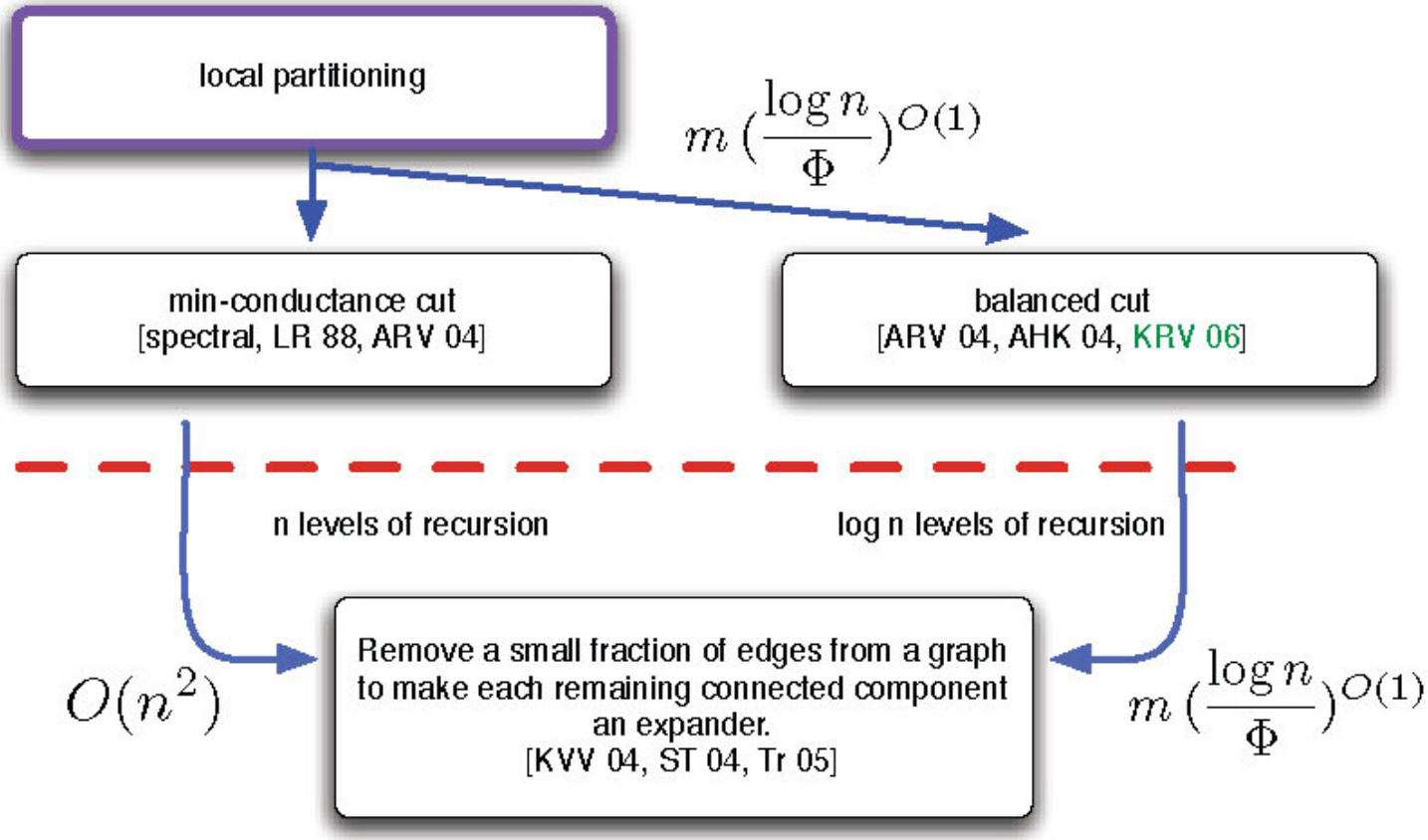
10x zoom



These pictures were made by applying local partitioning throughout the graph with random seed vertices and target

Theoretical applications

Local partitioning can be used as a subroutine to more quickly solve problems traditionally solved by recursive partitioning.



Four proofs of Cheeger inequalities



- graph spectral method
- random walks
- PageRank
- heat kernel

The Cheeger inequality $\Longleftrightarrow$ Partition algorithm

Using eigenvector f,

the Cheeger inequality can be stated as

$$2\Phi \geq \lambda \geq \frac{\alpha^2}{2} \geq \frac{\Phi^2}{2}$$

where λ is the first non-trivial eigenvalue of the Laplacian and α is the minimum Cheeger ratio in a sweep using the eigenvector f.

Proof of the Cheeger inequality:

$$\lambda_G \geq R(g_+)$$

from definition

$$= \frac{\sum_{u\sim v}(g_+(u)-g_+(v))^2}{\sum_u g_+^2(u)d_u}$$

$$= \frac{\left(\sum_{u\sim v}(g_+(u)-g_+(v))^2\right)\left(\sum_{u\sim v}(g_+(u)+g_+(v))^2\right)}{\sum_u g_+^2(u)d_u \sum_{u\sim v}(g_+(u)+g_+(v))^2}$$

$$\geq \frac{\left(\sum_{u\sim v}(g_+(u)^2-g_+(v)^2)\right)^2}{2\left(\sum_u g_+^2(u)d_u\right)^2}$$

by Cauchy-Schwarz ineq.

$$= \frac{\left(\sum_i |g_+(v_i)^2-g_+(v_{i+1})^2|\ \ |\partial(S_i)|\right)^2}{2\left(\sum_u g_+^2(u)d_u\right)^2}$$

$$\geq \frac{\left(\sum_i |g_+(v_i)^2-g_+(v_{i+1})^2|\alpha_G|\tilde{\text{vol}}(S_i)|\right)^2}{2\left(\sum_u g_+^2(u)d_u\right)^2}$$

from the definition.

$$= \frac{\alpha_G^2}{2}\frac{\left(\sum_i g_+(v_i)^2(|\tilde{\text{vol}}(S_i)-\tilde{\text{vol}}(S_{i+1})|)\right)^2}{\left(\sum_u g_+^2(u)d_u\right)^2}$$

summation by parts.

$$= \frac{\alpha_G^2}{2}\frac{\left(\sum_i g_+(v_i)^2 d_{v_i}\right)^2}{\left(\sum_u g_+^2(u)d_u\right)^2}$$

$$\alpha_G^2$$

Four proofs of Cheeger inequalities

- graph spectral method Fiedler '73, Cheeger, 60's
- random walks Lovasz, Simonovits, 90, 93
  Spielman, Teng, 04
- PageRank Andersen, Chung, Lang, 06
- heat kernel Chung, PNAS , 08.

A Cheeger inequality using random walks

Lovász, Simonovits, 90, 93

$$\left|W^k(u,S)-\pi(S)\right| \le \sqrt{\frac{vol(S)}{d_u}}\left(1-\frac{\beta_k^2}{8}\right)^k$$

Leads to a Cheeger inequality:

$$2\Phi \;\ge\; \lambda \;\ge\; \frac{\beta_G{}^2}{8} \;\ge\; \frac{\Phi^2}{8}$$

where β_G is the minimum Cheeger ratio over sweeps by using a lazy walk of k steps from every vertex for an appropriate range of k .

A Cheeger inequality using PageRank

Using the PageRank vector.

Recall the definition of PageRank $p=pr(\alpha,s)$:

(1) $$p = \alpha s + (1-\alpha) pW$$

(2) $$p = \alpha \sum_{t=0}^{\infty} (1-\alpha)^t (sW^t)$$

Organize the random walks by a scalar α.

A Cheeger inequality using PageRank

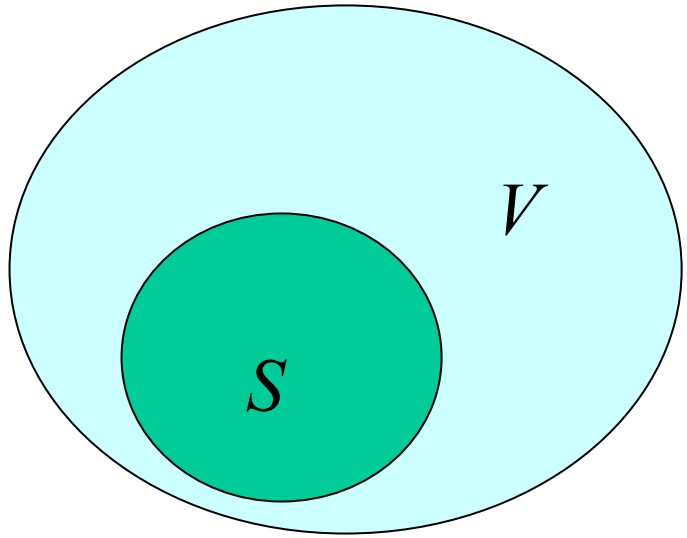
Using the PageRank vector with seed as a subset S and $vol(S) \leq vol(G)/4$, a Cheeger inequality can be obtained :

$$\Phi_S \geq \lambda_S \geq \frac{\gamma_S^2}{8 \log s} \geq \frac{\Phi_S^2}{8 \log s}$$

where $\bullet_S$ is the Dirichlet eigenvalue of the Laplacian, and γ_S is the minimum Cheeger ratio over sweeps by using personalized PageRank with seeds S.

Dirichlet eigenvalues for a subset $S \subseteq V$

$$\lambda_S = \inf_f \frac{\sum_{u:\, v} (f(u) - f(v))^2}{\sum_w f(w)^2 d_w}$$



over all f satisfying the Dirichlet boundary condition:

$$f(v) = 0 \qquad \text{for all } v \notin S.$$

A Cheeger inequality using PageRank

Using the PageRank vector with seed as a subset S and $vol(S) \leq vol(G)/4$, a Cheeger inequality can be obtained :

$$\Phi_S \geq \lambda_S \geq \frac{\gamma_S^2}{8 \log s} \geq \frac{\Phi_S^2}{8 \log s}$$

where $\bullet_S$ is the Dirichlet eigenvalue of the Laplacian, and γ_S is the minimum Cheeger ratio over sweeps by using personalized PageRank with seed S.

Algorithmic aspects of PageRank

- Fast approximation algorithm for personalized PageRank

  greedy type algorithm, linear complexity

- Can use the jumping constant to approximate PageRank with a support of the desired size.

- Errors can be effectively bounded.

A graph partition algorithm using PageRank

Given a set S with $\Phi_S \leq \alpha, vol(S) \leq \frac{1}{2} vol(G).$

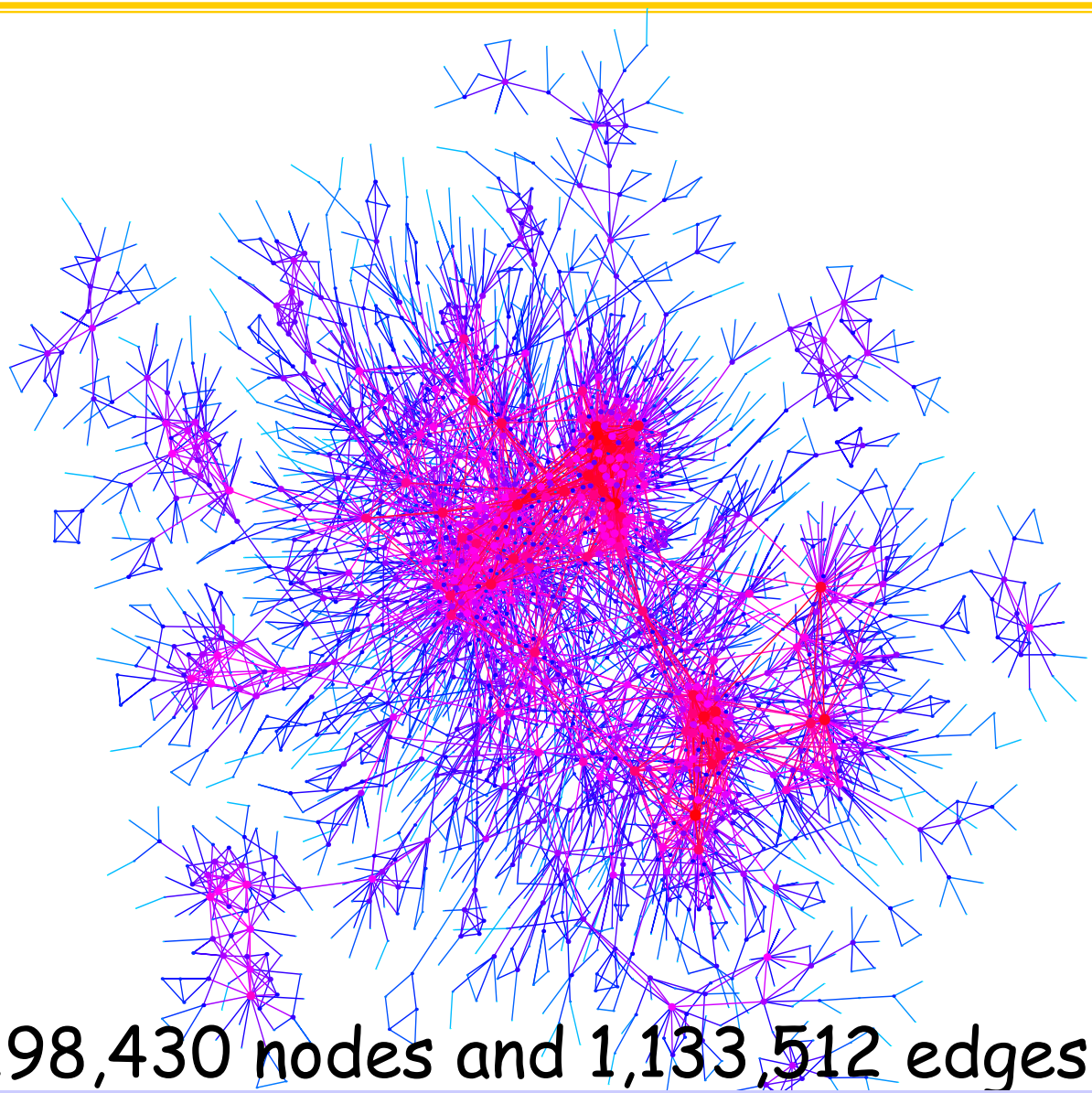
Randomly choose a vertex v in S.

With probability at least $\frac{1}{2}$,

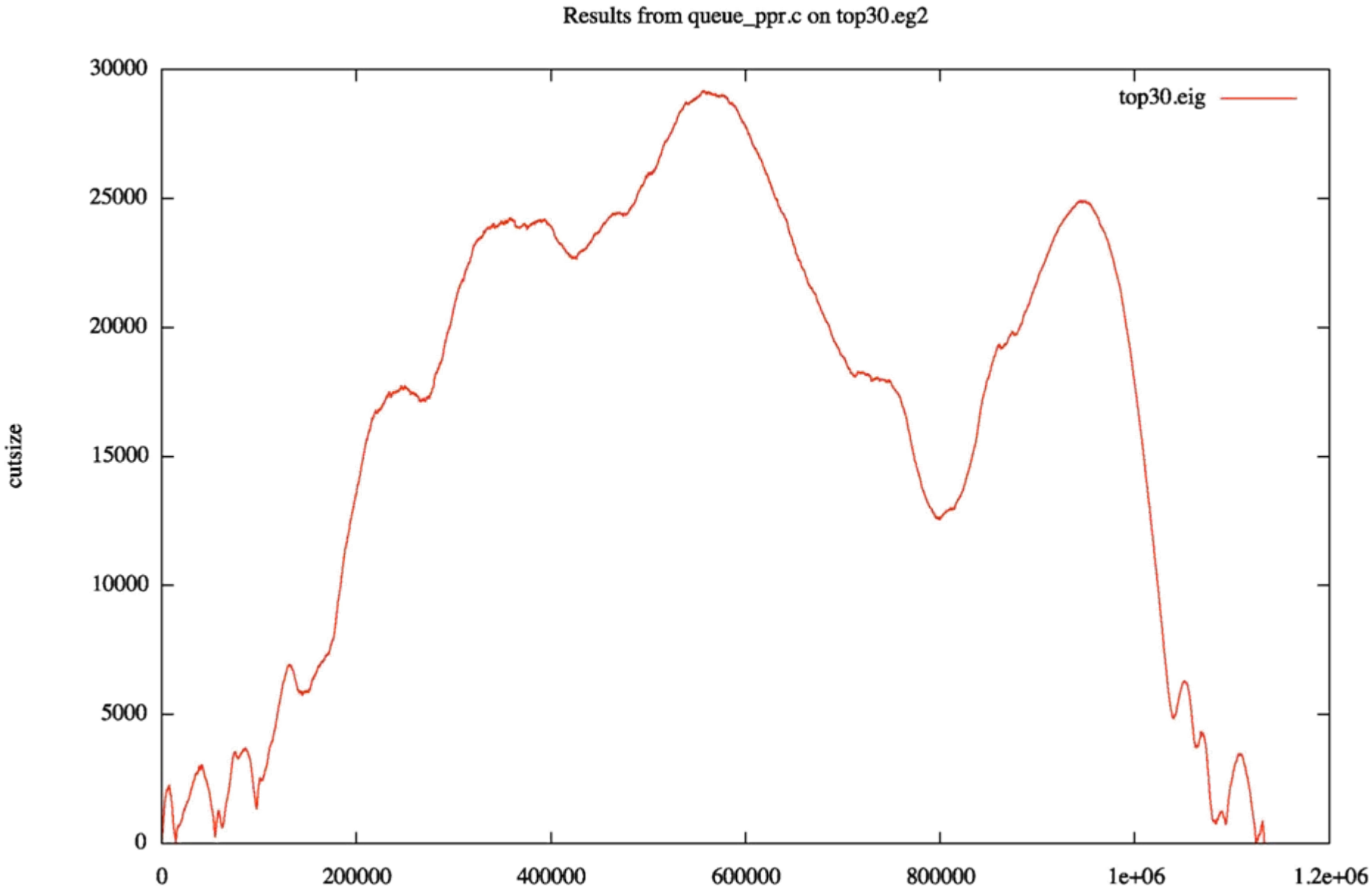
the one-sweep algorithm using $f = pr(\alpha, v)$

has an initial segment with the Cheeger constant at most $\Phi_f = O(\sqrt{\alpha \log |S|}).$

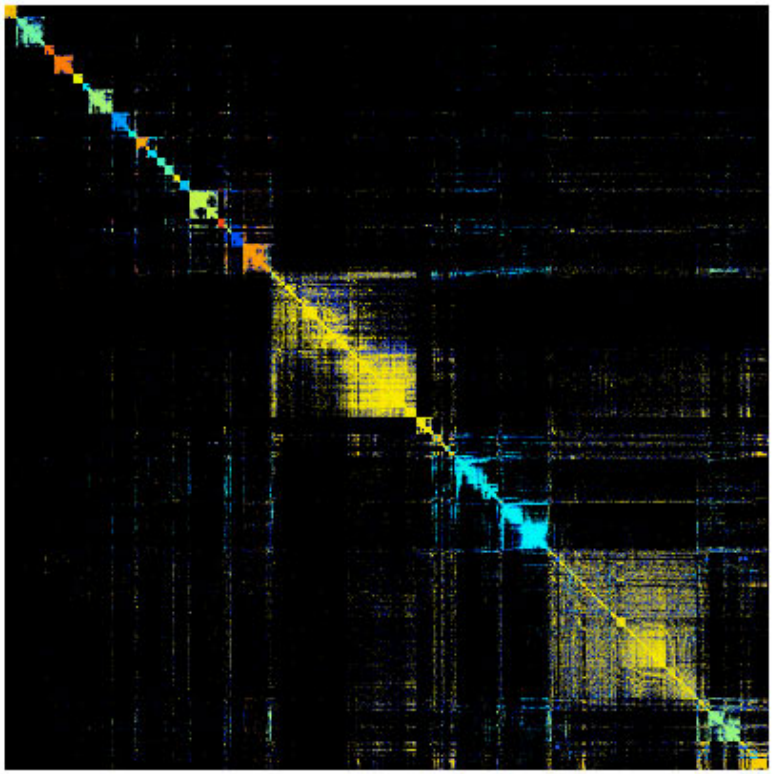
Graph partitioning using PageRank vector.



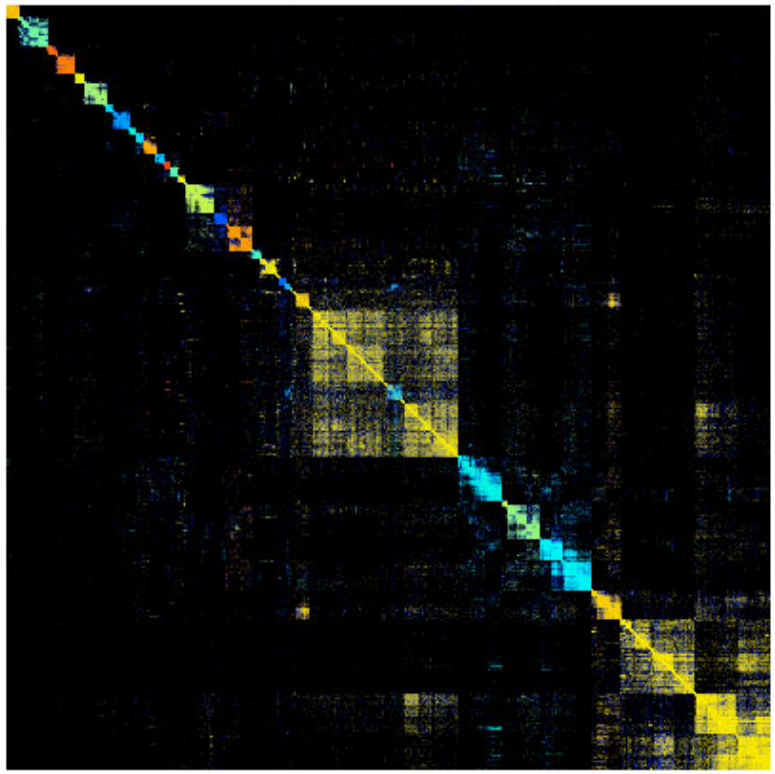
198,430 nodes and 1,133,512 edges


Results from queue_ppr.c on top30.eg2
top30.eig
cutsize
30000
25000
20000
15000
10000
5000
0
0
200000
400000
600000
800000
1e+06
1.2e+06

Internet Movie Database



Local partitioning
(10 min)



Recursive spectral partitioning
(250 min)

Four proofs of Cheeger inequalities

- graph spectral method Fiedler '73, Cheeger, 60's
- random walks Lovasz, Simonovits, 90, 93
  Spielman, Teng, 04
- PageRank Andersen, Chung, Lang, 06
- heat kernel Chung, PNAS , 08.

PageRank versus heat kernel

$$p_{\alpha,s} = \alpha \sum_{k=0}^{\infty} (1-\alpha)^k (sW^k)$$

Geometric sum

$$\rho_{t,s} = e^{-t} \sum_{k=0}^{\infty} s \frac{(tW)^k}{k!}$$

Exponential sum

$$p = \alpha + (1-\alpha) pW$$

recurrence



$$\frac{\partial \rho}{\partial t} = -\rho(I - W)$$

Heat equation

Definition of heat kernel

$$H_t = e^{-t}(I + tW + \frac{t^2}{2}W^2 + \ldots + \frac{t^k}{k!}W^k + \ldots)$$

$$= e^{-t(I-W)}$$

$$= e^{-tL}$$

$$= I - tL + \frac{t^2}{2}L^2 + \ldots + (-1)^k \frac{t^k}{k!}L^k + \ldots$$

$$\frac{\partial}{\partial t}H_t = -(I - W)H_t$$

$$\rho_{t,s} = sH_t$$

A Cheeger inequality using the heat kernel

Theorem:

$$\left|\rho_{t,u}(S)-\pi(S)\right| \le \sqrt{\frac{vol(S)}{d_u}} e^{-t\kappa_{t,u}^2/4}$$

where $\kappa_{t,u}$ is the minimum Cheeger ratio over sweeps by using heat kernel pagerank over all u in S.

Theorem: For $vol(S) \le vol(G)^{2/3}$,

$$\left|\rho_{t,S}(S)-\pi(S)\right| \ge \frac{e^{-t\lambda_S}}{2}.$$

A Cheeger inequality using the heat kernel

Using the upper and lower bounds,

a Cheeger inequality can be obtained :

$$\Phi_S \geq \lambda_S \geq \frac{\kappa_S^2}{8} \geq \frac{\Phi_S^2}{8}$$

where λ_S is the Dirichlet eigenvalue of the Laplacian, and κ_S is the minimum Cheeger ratio over sweeps by using heat kernel with seeds S for appropriate t.

Theorem:

$$\left|\rho_{t,S}(S)-\pi(S)\right| \geq (1-\pi(S))e^{-h_S t/(1-\pi(S))}$$

Sketch of a proof:

Consider $F(t) = -\log(\rho_{t,S}(S)-\pi(S))$

Show $\dfrac{\partial^2}{\partial t^2} F(t) \leq 0$

Then $\dfrac{\partial}{\partial t} F(t) \leq \dfrac{\partial}{\partial t} F(0) = \dfrac{\Phi_S}{1-\pi(S)}$

Solve and get $\left|\rho_{t,S}(S)-\pi(S)\right| \geq (1-\pi(S))e^{-h_S t/(1-\pi(S))}$

Random walks versus heat kernel

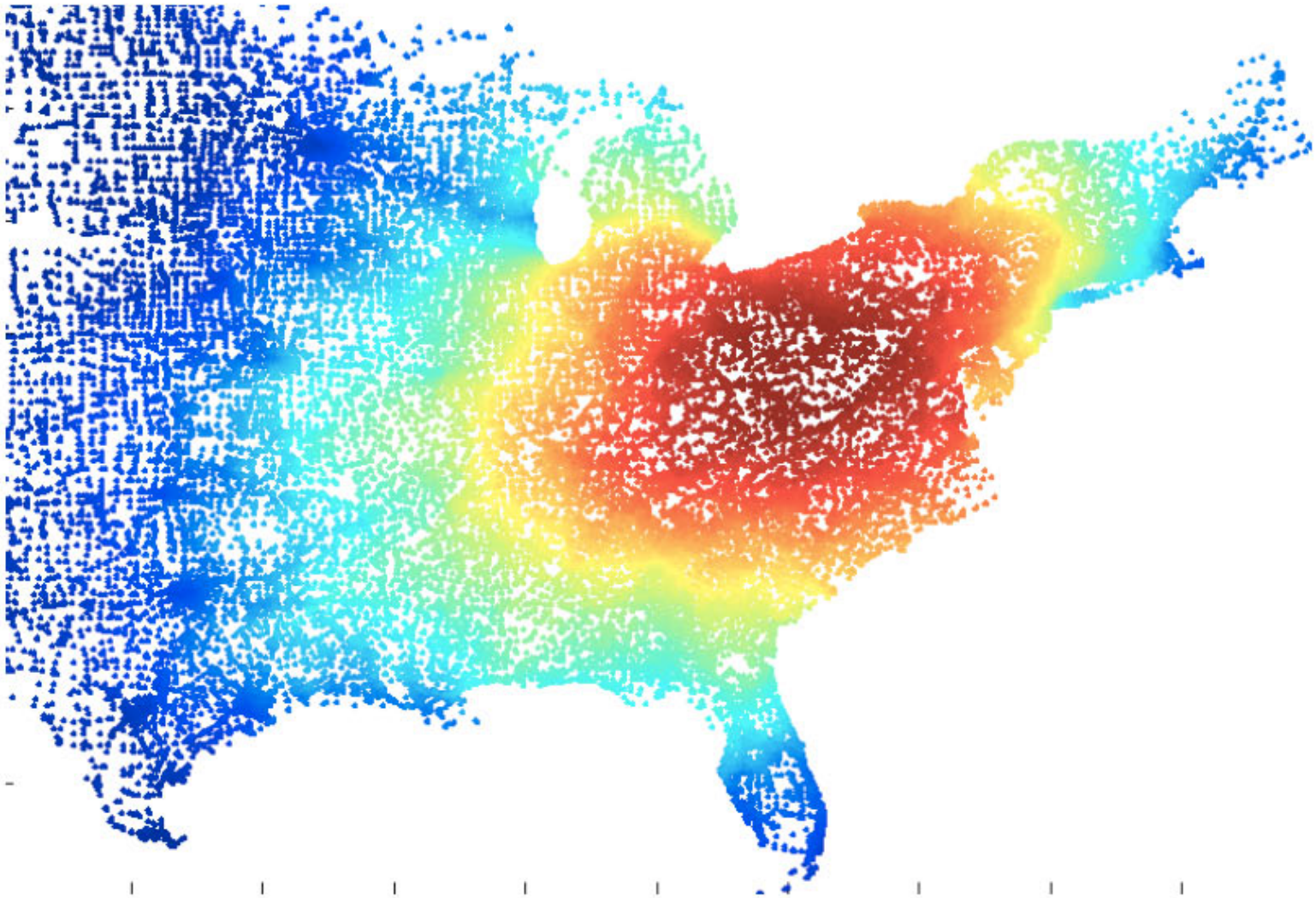
How fast is the convergence to the stationary distribution?

For what k, can one have

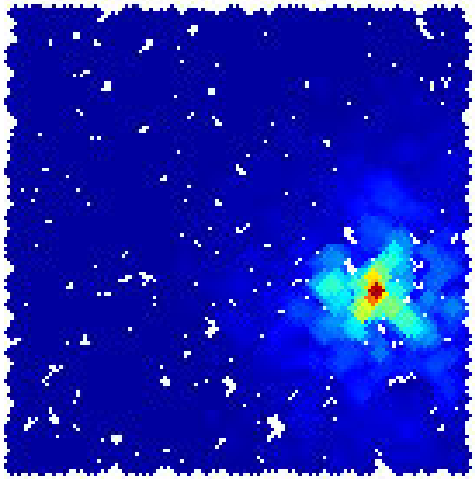
$$f W^k \to \pi \quad ?$$

Choose t to satisfy the required property.

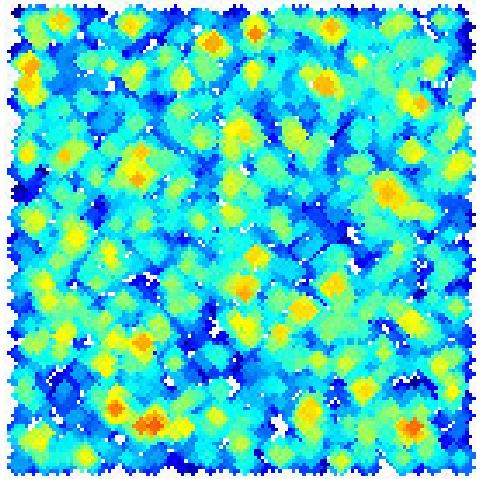
What the sweep should look like



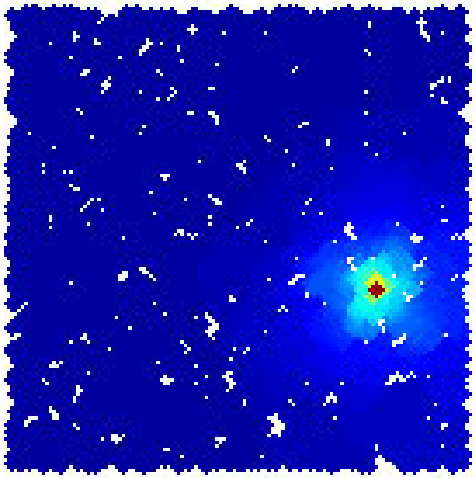
local pr



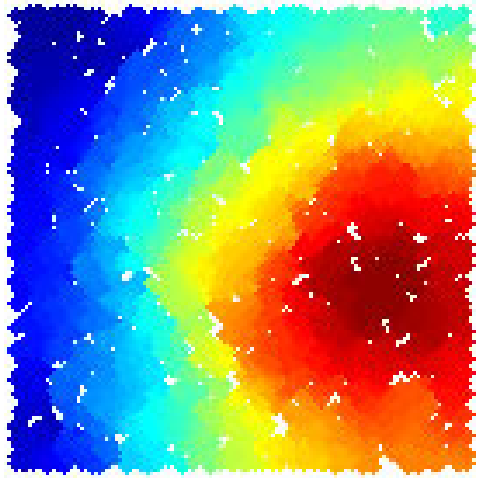
global pr

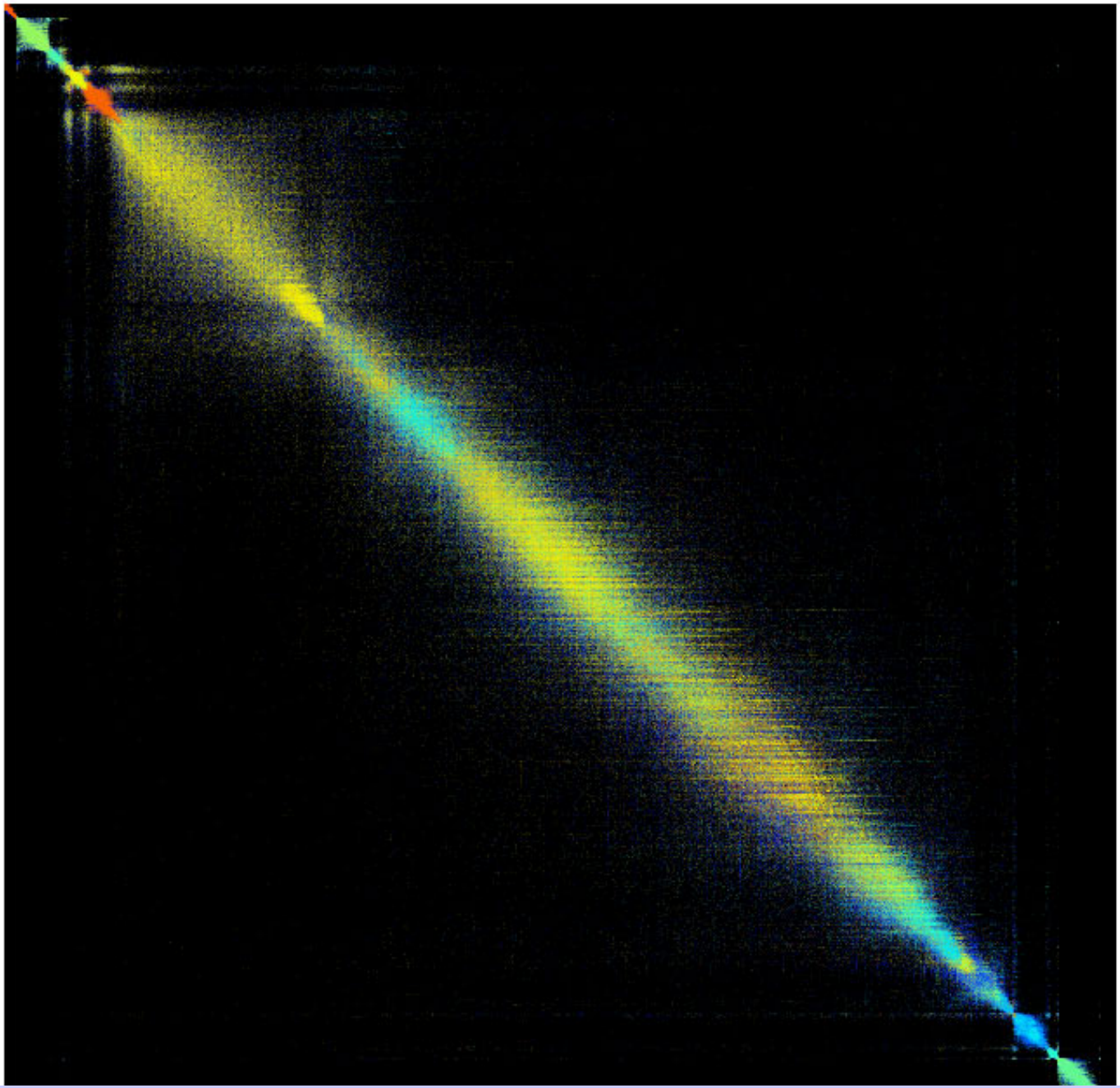


local/global ratio



sweep





New Directions in Graph Theory for information networks

Topics:
- Complex networks
- pageranks
- Games on graphs

Methods:
- Probabilistic methods
- Analytic methods

Related areas:
- Spectral graph theory
- Random walks
- Random graphs
- Game theory
- Quasi-randomness

Some related papers:

- Andersen, Chung, Lang, Local graph partitioning using pagerank vectors, FOCS 2006
- Andersen, Chung, Lang, Local partitioning for directed graphs using PageRank, WAW 2007
- Chung, Four proofs of the Cheeger inequality and graph partition algorithms, ICCM 2007
- Chung, The heat kernel as the pagerank of a graph, PNAS 2008.

